

PRD: Aster Roadside Assist

Vision and Goals

Aster Roadside Assist is an AI voice agent for the first minutes of a roadside assistance claim. It confirms safety, verifies identity, captures vehicle, location, and incident details, then gives the customer a clear next step without repeated handoffs.

The goal is better reassurance with tighter operating control: faster intake, more complete first-contact data, fewer manual coverage checks, fewer avoidable dispatch errors, lower cost-to-serve, and lower cost of revenue per case. Human agents stay focused on complex, high-risk, or low-confidence situations.

The prototype targets the objections insurers will raise before trusting automation: safety, vague locations, noisy incident descriptions, identity verification, privacy, coverage confidence, and human oversight. Success is measured by completeness, decision accuracy, escalation quality, auditability, dispatch waste, CSAT, and claim leakage.

Prioritized Features in the Prototype

The design answer is simple: the LLM handles conversation; a deterministic backend owns workflow state, validation, policy checks, and outcomes.

- 1 **Will the agent miss safety issues or create incomplete cases?** Intake starts with safety, then collects identity, vehicle, location, and incident step by step. The backend decides whether required facts are complete before coverage or next-best-action can run.
- 2 **Will the LLM improvise instead of following insurance rules?** Kotlin/Spring Boot owns validation gates, claim state, terminal outcomes, transcripts, and tool-call trace. The LLM can converse, but it cannot skip workflow requirements or silently decide coverage.
- 3 **Will it fail on vague roadside locations?** Backend geoparsing uses Google Maps links. If the first location is ambiguous, the agent asks a follow-up and the backend stitches caller clues into a dispatchable place. This proves the path for imperfect caller descriptions, not final production coverage.
- 4 **Will it respect privacy?** Customer and vehicle details are withheld from the conversational model until verification succeeds. Known callers provide selected PIN digits; unknown callers provide name, birthdate, and PIN digits. Failed verification stops intake without data disclosure.
- 5 **Will decisions be explainable to customers and agents?** A structured-output model classifies the incident into controlled types; backend policy logic maps it to covered, not covered, repair truck, tow truck, or human callback. The customer receives a fake SMS outcome.
- 6 **Will humans trust and audit it?** The observation UI shows case status, extracted facts, validation gates, transcript/tool trace, outcome, and KPIs.
- 7 **What happens when automation is wrong or uncomfortable?** The caller can ask for a human, and the system routes specialist, non-policyholder, or low-confidence cases to callback.

Milestones

This is scoped as an AI product, not a multi-year transformation programme, but the insurer still needs time to review safety, policy, and operational decisions. A strong FDE team with heavy AI assistance can build quickly; the plan uses weekly deliverables so the partner can decide at each checkpoint. Rollout comes after validation; direct dispatch automation, payment, policy changes, complex fraud handling, and full contact-centre migration stay out of scope until evidence supports them.

Week 1: Workflow contract, decision boundaries, and observer needs. Deliver safety/auth/location/incident/coverage flow, required fields, state transitions, escalation triggers, allowed/blocked AI actions, policy assumptions, 50-100 labelled eval calls, and the insurer's required observer views/KPIs. End-of-week check-in: ops/compliance signs off what AI may ask, what it may never decide, required pre-dispatch facts, and supervisor visibility.

Week 2: Backend authority and real data paths. Build Spring-owned claim state, PIN verification, real policy/user/customer-data connectors where available, workflow integration, validation gates, incident enums, coverage rules, terminal outcomes, durable transcript storage, and tool-call audit. End-of-week check-in: real demo shows intake against connected data and proves the LLM cannot skip auth, safety, location, incident, or policy gates.

Week 3: Matching quality and controllable voice stack. Improve geoparsing for motorway direction, junction ranges, service stations, landmarks, repeated clues, confidence scoring, Google Maps links, and "ask again" behavior. Replace the Realtime-only demo with a controllable STT + LLM + TTS path. Verify incident and location matching against historical recordings if available. End-of-week check-in: ambiguous road and incident evals return dispatchable matches or clear follow-ups.

Week 4: Operational integrations and final observer. Connect the voice stack to a real phone line, SMS line, dispatch services for tow/repair trucks, and human callback services. Implement the insurer's observer wishes from Week 1 and finalize status, facts, transcript, tool trace, outcome, SMS, and KPIs. End-of-week check-in: final voice model/stack and observation UI handle scripted calls with one clear next step or callback route.

Week 5: Shadow validation and rollout readiness. Run the built flow beside agents or replay calls. Compare AI output against human intake for completeness, safety, identity, location, incident, escalation, latency, and tone. End-of-week check-in: no unacceptable safety/privacy/policy failures, signed issue list, and agreed rollout criteria.

+ **Rollout.** After the build is validated, rollout starts with a narrow signed-off slice and expands only from evidence. Each expansion needs its own eval set, pilot threshold, rollback plan, and human-control boundary.

Technical Risks and Open Questions

Location resolution. Google Maps works for many landmarks and road descriptions, but roadside cases may require road-network data, highway marker handling, "between junctions" parsing, and provider coverage zones. The prototype proves potential, not production completeness.

Voice reliability. Realtime voice is fast to prototype, but production needs measured reliability for noise, VAD, interruptions, call ending, roadside STT, TTS pacing/tone, road names, junctions, number plates, PIN digits, and vehicle terms. Splitting STT + LLM + TTS gives more control while orchestration, state, tool policy, audit trail, and routing stay under our control.

Policy and workflow. Production needs a clearer joint view of the client's roadside workflow: what agents ask, which systems they check, which decisions are rules vs. judgement calls, when they escalate, and how dispatch partners are selected. Only then should we encode policy rules, state steps, and handoff points. The LLM should classify and extract facts, not invent process.

Privacy and security. Production needs stronger authentication choices, consent and recording disclosures, PII minimization, retention rules, fraud monitoring, and full audit logging.

AI Integration Approach

The system separates conversational AI from business authority. The initial demo can use OpenAI Realtime for speed, but the production-shaped build moves to a controllable STT + LLM + TTS stack plus a structured-output model to classify incidents into controlled enums. The backend state machine determines the next required fact and whether the case can proceed. A backend geoparsing service resolves spoken locations into dispatchable places. Coverage and next-best-action decisions are backend-owned and traceable.

The improvement loop is eval-driven before and after launch. Pre-launch, historical recordings and transcripts seed evals for geoparsing, incident detection, prompts, state-machine changes, and STT/TTS choices. QA shadow mode means the AI runs beside real or replayed roadside calls, proposes extracted facts, dispatchable location, incident type, coverage readiness, and handoff/next step, while humans still own the customer outcome. QA labels mismatches against the human record. Post-launch, every call should produce audio, transcript, corrected transcript, extracted facts, tool trace, decision path, and verified outcome; those labels improve location matching, incident classification, follow-up questions, and escalation quality. Changes promote only when thresholds improve.